6-DOF Parallel Kinematic Robot for Respiratory Motion Simulation in Human Anthropomorphic Phantoms

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Abstract

This study presents the mechanical architecture and analytical framework of a six-degree-of-freedom (6-DOF) Stewart–Gough parallel kinematic mechanism for respiratory motion simulation using an anthropomorphic abdominal phantom. Six independently driven linear actuators connect a fixed base to a moving phantom-support platform. The 341-component prototype comprises 108 bearings, custom universal joints and three-dimensionally inclined actuator mounting wedges. The platform is intended to reproduce translational and rotational components of respiratory motion for moving-phantom experiments involving surgical instrument positioning and respiratory-motion compensation. Actuator-length inverse kinematics, the differential Jacobian, elementary loading relations and joint-space constraints are derived. Periodic respiratory waveforms are developed with reference to published models, followed by a methodology for patient-specific reference generation and phantom-based experimental validation.

Keywords

Stewart–Gough platform; 6-UPS; parallel manipulator; inverse kinematics; respiratory motion simulation; anthropomorphic phantom; motion compensation; robotic needle insertion; patient-specific motion profiles.

1. Introduction and rationale

Breathing generates quasi-periodic displacement of the chest wall, diaphragm and abdominal structures. During image-guided procedures, internal targets can move relative to the instrument frame, and changes in breathing phase may alter the effective target position. A bench-top motion platform provides a repeatable way to study these effects without beginning with live subjects. A six-axis parallel manipulator offers simultaneous vertical, lateral, longitudinal and small-angle rotational motions, allowing prescribed rigid-body motion traces to be imposed beneath an abdominal phantom.

The intended simulator is not a complete model of tissue biomechanics. A rigid Stewart platform controls pose of its moving plate; displacement of a simulated lesion within an abdomen depends on phantom geometry, mounting stiffness and internal tissue analogues. In consequence, reproducing an external abdomen trajectory alone must not be interpreted as reproducing internal tumor motion until a mapping and measurement procedure is established.

Clinical investigations document anisotropic tumor motion and inspiration–expiration asymmetry [3], while the AAPM task-group report emphasizes patient-specific motion measurement when appropriate [4]. These clinical references motivate, but do not validate, the present prototype.

1.1 Project objective

To create an instrumented, programmable six-actuator motion base that supports an anthropomorphic abdominal phantom and reproduces prescribed 6-DOF respiratory poses, permitting quantified investigations of targeting errors, motion prediction and compensation in a controlled laboratory environment.

1.2 Technical questions

- How is an anatomical 6-DOF displacement transformed into six actuator-extension commands while respecting platform geometry?
- Which respiratory waveforms are analytically suitable as controllable references, and how should patient-derived traces be incorporated without imposing unsupported physiological assumptions?
- How do link orientation, universal-joint limits, actuator stroke and Jacobian conditioning bound the feasible respiratory workspace?
- What measurements establish whether the physical phantom follows the commanded motion and whether a target inside it follows the platform pose?

2. Related research and distinction from prior art

Dasgupta and Mruthyunjaya [1] review Stewart-platform configuration, analysis and design, including the strong coupling of pose and actuator lengths and the importance of singularity assessment. Their review provides the theoretical foundation for the geometry and differential motion relations in Section 4. Lujan et al. [2] introduce an asymmetric periodic representation of organ motion in a dose-calculation context; it is used here solely as a reference waveform family, not as evidence that any particular patient breathes to an ideal curve.

Seppenwoolde et al. [3] report direct three-dimensional tumor-motion measurements and demonstrate nonuniform motion with respiratory asymmetry and hysteresis. These observations justify separate directions, phase offsets and patient-specific reconstruction. AAPM Task Group 76 [4] provides a wider context for motion measurement and management in radiotherapy.

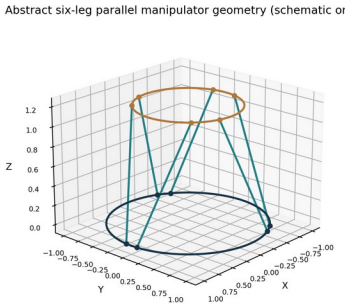
Especially relevant existing prior art is the study by Kim et al. [5], which uses real respiratory-motion data, a Stewart–Gough platform and a gelatin phantom in a robot-assisted needle-insertion investigation. Gunderman et al. [6] report a respiratory motion-compensated parallel robotic system for abdominal ablation. These works establish that the general concept of using a Stewart platform to reproduce respiratory motion for robotic needle tasks is not, by itself, new; potential intellectual-property claims require narrower, verifiable constructional or control distinctions and a formal prior-art examination.

3. System concept and physical assembly

3.1 Mechanism architecture

The mechanism consists of a stationary base, six extensible legs and a mobile plate supporting the phantom. Each actuated leg transmits motion between the base and upper platform through articulated joints. A 6-UPS (universal–prismatic–spherical)

topology provides the nominal kinematic model; joint configuration and attachment geometry in the physical prototype must be confirmed using final assembly measurements. Each leg is modeled as a length-constrained link between corresponding fixed and mobile anchor points.



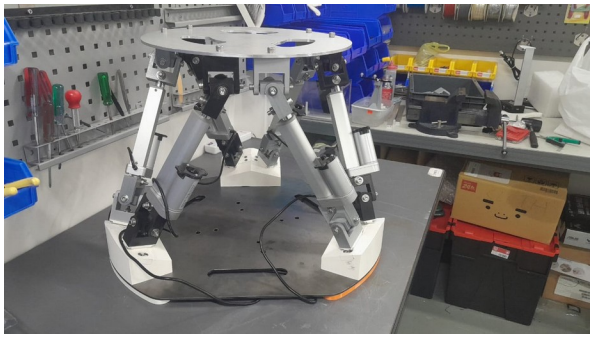


Figure 2. Original project overview of the Stewart-platform respiratory simulator.

Figure 1. Geometric interpretation of six legs between fixed anchors and a moving upper plate (author-generated schematic, not to scale and not a reconstruction of actual CAD dimensions).

3.2 Hardware composition and fabrication record

The constructed prototype contains 341 components, including 108 bearings (54 radial and 54 thrust), with approximately 420 screw-fastening operations during assembly. The custom universal joints provide multi-axis articulation within the actuator installation envelope. Tri-axis inclined wedge mounts position the actuators to accommodate the intended motion range and reduce interference. Their effect on usable workspace and singularity proximity requires quantitative evaluation using the final anchor coordinates and joint limits.

Stationary frame — Base plate, inclined leg supports, custom wedges [Engineering consideration: Rigidity; anchor calibration; fixture mounting]



Figure 3. CAD assembly of the six-actuator parallel kinematic platform.

Moving plate — Circular top plate / anthropomorphic-phantom support [Engineering consideration: Mass distribution; accessible fixing points]

Actuation — Six independent linear actuator assemblies [Engineering consideration: Stroke, velocity and thrust margins]

End joints — Custom U-joints; 108 total radial/thrust bearings [Engineering consideration: Angular clearance; assembly tolerance; friction]

Sensing and control — Proposed STM32 controller and encoder-based inverse-kinematics control [Requires calibration and validation]

Safety interlocks — Limit switches and reference positions described in project concept [Engineering consideration: Document actual wiring and stopping distances]

3.3 Original mounting and phantom integration

The source presentation includes a CAD visualization of a container/abdominal-phantom surrogate placed above the moving plate. The top plate serves as the mechanical interface from platform pose to phantom displacement. To enable realistic surgical-target experiments, the external phantom fixture must be designed for a defined center of mass, mounting repeatability and measurable internal targets. Registration markers should distinguish base coordinates, platform coordinates and internal target coordinates.

4. Kinematic formulation for the six-actuator platform



Figure 4. Photograph of the assembled six-actuator prototype.

4.1 Reference frames and symbols

Define frame {B} at the fixed base and frame {P} at the movable plate. Let $b_i \in \mathbb{R}^3$ be the measured location of base anchor i in {B}, and $a_i \in \mathbb{R}^3$ the corresponding platform anchor in {P}, for i = 1,…,6. Let $p = [x,y,z]^T$ be the platform-origin position in {B}, and $R \in SO(3)$ the orientation transforming platform-frame vectors into base-frame vectors. For consistency with typical robotics conventions, take $R = Rz(\psi)\, Ry(\theta)\, Rx(\varphi)$, representing yaw, pitch and roll about the stated axes; other rotation conventions require changing the formulas accordingly.

b_i, a_i — Fixed-base / moving-plate attachment-point vectors (mm)

p, R — Moving-plate translation vector / 3 × 3 rotation matrix

φ, θ, ψ — Roll, pitch, yaw angles (rad unless otherwise stated)

s_i, ℓ_i, u_i — Leg vector, leg length and corresponding unit vector

J — 6 × 6 velocity Jacobian

q(t) — Six-component commanded platform pose over time

A, T, n — Motion amplitude (mm), breathing period (s), shape exponent

4.2 Closed-form inverse position kinematics

For a prescribed platform pose, the base-frame location of the upper joint is $p + R\, a_i$. The i-th strut is the vector from its base anchor to this upper attachment. Its commanded joint-center separation is therefore the Euclidean norm of that vector. This calculation is repeated independently for all six actuators; unlike the direct pose solution, standard actuator-length inverse kinematics are algebraically straightforward once anchor coordinates are known [1].

$$s_i = p + R\, a_i - b_i, \quad i = 1,\ldots,6 \quad (1)$$

$$\ell_i(p, R) = \sqrt{(s_i^T s_i)} = \|p + R\, a_i - b_i\|_2 \quad (2)$$

$$\Delta\ell_i(t) = \ell_i[p(t), R(t)] - \ell_{i,\mathrm{ref}} \quad (3)$$

Here $\ell_{i,\mathrm{ref}}$ denotes the calibrated length at a reference pose, and $\Delta\ell_i$ is the actuator displacement from that reference. To apply Equation (2) to the prototype, the six base and platform attachment coordinates must first be established from the final CAD geometry or calibrated measurements.

4.3 Orientation representation

Using the intrinsic yaw–pitch–roll ordering above, rotation is constructed by multiplication of orthogonal axis-rotation matrices. Near small respiratory angles the rotation may be linearized, but the full nonlinear expression should be used in a final command generator if significant rotational motion is prescribed.

$$R = Rz(\psi)\, Ry(\theta)\, Rx(\varphi) \quad (4)$$

$$Rx(\varphi) = [[1,0,0],[0,\cos\varphi,-\sin\varphi],[0,\sin\varphi,\cos\varphi]] \quad (5)$$

$$Ry(\theta) = [[\cos\theta,0,\sin\theta],[0,1,0],[-\sin\theta,0,\cos\theta]] \quad (6)$$

$$Rz(\psi) = [[\cos\psi,-\sin\psi,0],[\sin\psi,\cos\psi,0],[0,0,1]] \quad (7)$$

4.4 Differential kinematics and Jacobian

Differentiating the leg vector gives $\dot{s}_i = \dot{p} + \omega \times (R\, a_i)$, where ω is the platform angular velocity expressed in {B}. With the leg unit vector $u_i = s_i / \ell_i$, differentiating the norm yields a scalar extension-rate relation. Stacking the six scalar relations yields a velocity Jacobian mapping the six-component platform twist to the six actuator velocities.

$$\dot{\ell}_i = u_i^T \dot{p} + [\,(R\, a_i) \times u_i\,]^T \omega \quad (8)$$

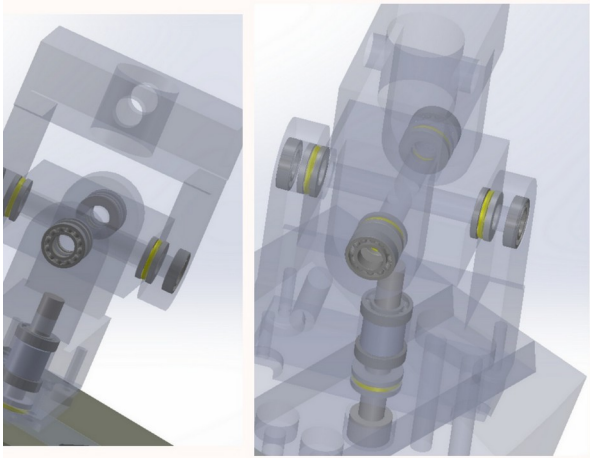


Figure 5. CAD detail of the custom universal-joint and bearing assembly.

$$\dot{\ell} = J\, [\,\dot{p}^T \;\; \omega^T\,]^T \quad (9)$$

$$J_{i,:} = [\,u_i^T \;\; ((R\, a_i) \times u_i)^T\,] \quad (10)$$

The rotational part of J has different physical units from the translational part; a normalized Jacobian, with a declared characteristic length, is preferable to an unscaled numerical condition number when comparing poses. Small singular values identify approaching loss of controllability or force amplification. A command path should therefore be checked against the actuator stroke, maximum leg rate, universal-joint articulation, self-collision and normalized Jacobian condition before being sent to hardware.

4.5 Quasi-static wrench balance and actuator loading

Ignoring joint friction, leg inertia and compliance at first approximation, the sum of axial leg-force contributions must balance an external wrench acting on the platform. In this common idealization the generalized wrench is the transpose of the kinematic Jacobian multiplied by the six leg forces. Dynamic design should extend this relation to gravity, payload inertia, link inertia and nonlinear losses.

$$w = J^{T} f, \quad \text{with } w = [F^{T}\ \tau^{T}]^{T} \text{ and } f = [f_1,\ldots,f_6]^{T} \quad (11)$$

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = Q + w_{ext} \quad \text{(general dynamic structure)} \quad (12)$$

Equation (12) describes the general dynamic formulation rather than a fully identified model of the prototype. Inertia, transmission efficiency, friction and center-of-mass parameters require experimental determination.

4.6 Assembly constraints and calibration

- Determine all twelve joint-center coordinates in one common coordinate convention, preferably exported directly from the final CAD assembly and independently checked by measurement.
- Calibrate encoder counts to physical extension, accounting for zero offset, backlash, load-dependent deflection and limit-switch offsets.
- Check each custom U-joint against its available articulation and the orientations commanded by the prescribed pose. The presentation illustrates approximately 40° motion for a joint configuration, but this is not a verified all-direction workspace limit.
- Validate the reference moving-plate pose by measured platform markers rather than assuming geometric symmetry or nominal bolt coordinates.
- Include out-of-range handling for commanded leg length, velocity, acceleration, synchronization error and approaching singularities.

5. Respiratory motion: literature-supported waveform models

5.1 Why a pure sine wave is insufficient

A sinusoidal vertical displacement is a useful first commissioning reference because its velocity and acceleration are smooth and easily calculated. Real respiratory behavior can differ: the inhale and exhale portions may be unequal, baseline and amplitude can drift, and three-dimensional internal tissue trajectories can exhibit hysteresis [2–4]. Thus a well-controlled platform should accept several classes of inputs rather than be hard-coded to a single sinusoidal amplitude.

5.2 Symmetric and asymmetric analytic signals

For initial testing, let the dimensionless periodic inspiration signal s(t) range from zero to one with period T. A symmetric reference and a $\cos^{2n}$-style asymmetric reference can be written as follows. Lujan et al. [2] used an asymmetric periodic approximation in the context of liver motion. In this report n = 2 is chosen as a visual illustration rather than fitted to a measured patient.

$$s_{sin}(t) = \tfrac{1}{2}[1 + \cos(2\pi t/T + \phi_0)] \quad (13)$$

$$s_{asym}(t) = \{\tfrac{1}{2}[1 + \cos(2\pi t/T + \phi_0)]\}^{n} = \cos^{2n}(\pi t/T + \phi_0/2) \quad (14)$$

$$d_z(t) = z_0 + A_z\, s(t) \quad (15)$$

The asymmetric form produces a comparatively brief inspiratory peak and broader low-displacement interval in the selected phase convention. The corresponding displacement physically reflects inhale or exhale only after the coordinate sign, phase and physiological interpretation have been established. The plot below is generated directly from Equations (13)–(14) with T = 4 s, n = 2 and normalized amplitude.

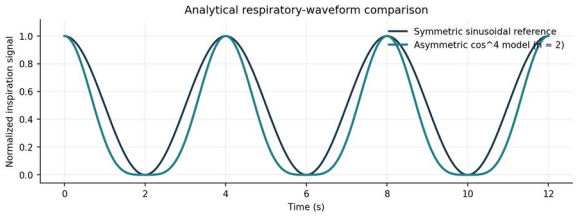


Figure 7. Research-informed analytical breathing curves: symmetric reference and a Lujan-style asymmetric waveform (author-generated plotting of mathematical functions, not copied or digitized patient data).

5.3 Nonstationary or patient-specific profiles

Patient-specific respiratory input is represented by a measured displacement time series d(t) rather than a single universal analytic function. Candidate acquisition methods include synchronized motion-capture markers, external respiratory tracking, optical surface measurement and clinically appropriate positional sensors. A reproducible patient-specific implementation requires a defined sampling protocol, signal processing and synchronized validation against the measured target motion.

A general experimental signal is $d(t) = b(t) + A(t)\cdot s[\phi(t)] + \varepsilon(t)$, where b(t) is baseline drift, A(t) is a time-dependent amplitude, $\phi(t)$ is respiratory phase and $\varepsilon(t)$ is measurement noise or higher-frequency residual motion. Filtering must preserve clinically relevant peaks and phase timing. Splitting training and test segments by patient or session is necessary if prediction methods are later compared.

$$d(t) = b(t) + A(t)s[\phi(t)] + \varepsilon(t) \quad (16)$$

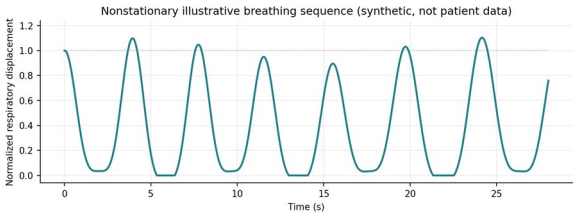


Figure 8. Example of phase variation and baseline drift in a synthetic respiratory trace; generated for controller-input illustration only, not derived from any patient.

5.4 Six-degree-of-freedom motion assignment

An abdominal surface trajectory can be converted into a six-component platform reference $q(t) = [x(t),y(t),z(t),\varphi(t),\theta(t),\psi(t)]^{T}$. For preliminary calibration, each coordinate may be defined as an amplitude-scaled, phase-shifted copy of a common waveform. This is a tunable kinematic representation, not a physiological equivalence; actual abdominal deformation and internal target displacement are not generally rigid-body transformations.

$$q_k(t) = q_{k,0} + A_k\, s(t - \delta_k), \quad k = 1,\ldots,6 \quad (17)$$

The six amplitudes A_k and phase lags δ_k should be measured or fitted to an experimentally defined phantom/target mapping. For a clinically motivated tumor trajectory, the coordinates and amplitudes must be chosen with regard to the anatomy under investigation. Published lung-tumor statistics [3] should not be applied indiscriminately to an abdominal phantom or to a different organ.

5.5 Internal-target mapping and hysteresis

If an internal target marker is present, let its base-frame position be c_B(t). A strictly rigid assumption predicts c_B(t) = p(t) + R(t)c_P. A deformable phantom may instead require c_B(t) = p(t) + R(t)c_P + δ_def(t), where δ_def represents target motion relative to the local phantom frame. Hysteresis may make δ_def depend on the direction of breathing as well as instantaneous displacement. Target motion should therefore be validated separately from top-plate pose before the system is used to judge a needle-positioning algorithm.

$$c_B(t) = p(t) + R(t)c_P + \delta_{def}(t) \quad (18)$$

6. Proposed controller and software implementation

6.1 Functional command pipeline

The control sequence maps an analytic or measured respiratory reference to a feasible platform trajectory, computes the corresponding actuator setpoints using Equation (2), and commands six synchronized feedback loops. Encoder measurements, calibrated zero positions, actuator travel bounds and supervisory fault monitoring are required for closed-loop operation. An STM32-based controller with encoder feedback and limit switches is a proposed implementation architecture; completed respiratory-motion tracking performance must be established experimentally.



Figure 9. Proposed control architecture from breathing reference to measured moving-phantom response (author-generated block diagram).

6.2 Joint-space tracking control

For each actuator i, define $e_i(t) = \Delta\ell_{i,des}(t) - \Delta\ell_{i,meas}(t)$. A baseline servo may use a velocity feedforward term with bounded proportional–integral–derivative action, actuator saturation and anti-windup. Gain selection must reflect motor drive hardware, sample frequency, position-encoder resolution and payload behavior. Full-platform Cartesian feedback additionally requires independent pose measurement or an appropriately qualified forward-pose estimator.

$$e_i(t) = \Delta\ell_{i,des}(t) - \Delta\ell_{i,meas}(t) \quad (19)$$

$$u_i(t) = K_{p,i}\, e_i + K_{i,i}\int e_i dt + K_{d,i}\, d\, e_i/dt + u_{ff,i} \quad \text{(illustrative)} \quad (20)$$

6.3 Forward pose estimation

A numerical forward-kinematics estimate can be obtained by minimizing the squared residual between measured actuator lengths and the lengths predicted by Equation (2). This is a nonlinear least-squares optimization over platform translation and rotation, initialized near the last valid pose. Because direct kinematics may admit multiple mathematical solutions, a warm start and independent motion capture are recommended for commissioning.

$$\hat{q} = \arg\min_q \Sigma_i[\ell_{i,meas} - \ell_i(q)]^2 \quad (21)$$

6.4 Command integrity and safety limits

- Software interlock: reject any pose violating calibrated minimum/maximum stroke, universal-joint limit, workspace boundary or link/fixture collision envelope.
- Trajectory interlock: bound leg speed and acceleration and maintain synchronized timing for all six legs.
- Fault detection: detect encoder disconnects, inconsistent leg errors, actuator stalls, watchdog expiry and home-switch disagreement.
- Test governance: use the system only with laboratory phantoms until appropriate engineering, regulatory and clinical validation has been completed.

7. Illustrative numerical kinematics example

An illustrative numerical example demonstrates conversion of a respiratory reference signal into six actuator trajectories. A nominal geometry is used because final attachment coordinates have not been calibrated. Consequently, these results demonstrate the computational method rather than measured performance of the physical prototype.

Base anchor radius — 170 mm (hypothetical)

Moving-plate anchor radius — 114 mm (hypothetical)

Neutral plate height — 235 mm (hypothetical)

Vertical breathing excursion — 8 mm peak-to-trough, imposed analytical input

Additional pose variations — Small x/y shifts and up to ~1° rotations; illustrative

Reference waveform — 4-s cycle, asymmetric n = 2; three cycles plotted

Data status — Synthetic geometry and motion; not a hardware or patient measurement

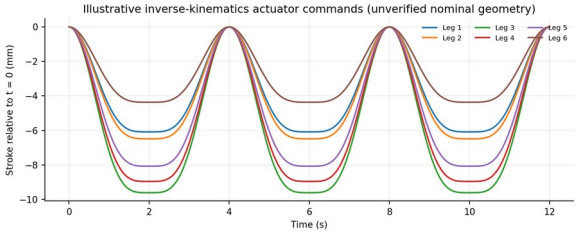


Figure 10. Example leg-extension curves obtained with Equation (2) using the explicitly hypothetical anchor geometry above. Curves are synthetic references, not actuator-encoder recordings.

This example demonstrates that a single vertical waveform does not generally yield six identical leg commands: asymmetrical attachment geometry and imposed platform rotations cause leg-specific extensions. The plot is included to show algorithm behavior and chart conventions, not the calibrated workspace of the physical machine.

8. Experimental methodology and validation plan

8.1 Mechanical geometry verification

Extract base and moving-plate joint-center coordinates from the assembled CAD model, then cross-check against a coordinate measurement of the prototype. Confirm all actuator stroke limits, actual zero positions, the custom joint angular envelope and potential interference with the phantom or neighboring legs. Record each source of tolerance in a configuration-controlled assembly and calibration sheet.

8.2 Static and dynamic response testing

Command a grid of static Cartesian poses spanning the intended respiratory motion envelope and record their realized translations and rotations using an external camera-based tracking system, laser displacement sensing, a calibrated 6-DOF tracker or other appropriate metrology. Repeat each pose multiple times to determine bias and repeatability. For dynamic testing, use regular sinusoidal and asymmetric waveforms, then time-stamped recorded respiratory inputs, with all measurement streams synchronized.

8.3 Proposed performance metrics

Pose RMSE — Root mean square difference between commanded and externally measured translation/orientation [Interpretation: Spatial tracking fidelity]

Peak error — Maximum absolute tracking deviation during a specified test interval [Interpretation: Worst observed excursion]

Temporal lag — Delay from cross-correlation or phase of output versus reference [Interpretation: Breath-cycle synchronization]

Repeatability — Standard deviation over repeated nominally identical cycles [Interpretation: Mechanical/control variability]

Actuator tracking — Encoder set-point minus actual encoder position [Interpretation: Joint-space tracking]

Phantom target error — Internal target-marker displacement versus prescribed target curve [Interpretation: Suitability for surgical studies]

Cycle endurance — Change in performance after repeated loaded cycles [Interpretation: Durability and drift]

8.4 Phantom-Based Needle Insertion Experiment

As a separate follow-on research study, a robotic arm equipped with a remote-center-of-motion needle-insertion end effector can act on a target inside the breathing phantom. A baseline insertion can first be conducted with the phantom stationary, followed by the same intervention under identical programmed respiratory motion with and without algorithmic motion compensation. Outcomes should include target miss distance, insertion depth error, tissue-analogue deformation, collision clearance and operation time. Such a study must use instrumented phantoms and properly reviewed protocols; a surgical outcome must not be claimed from platform-motion accuracy alone.

8.5 Example verification sequence

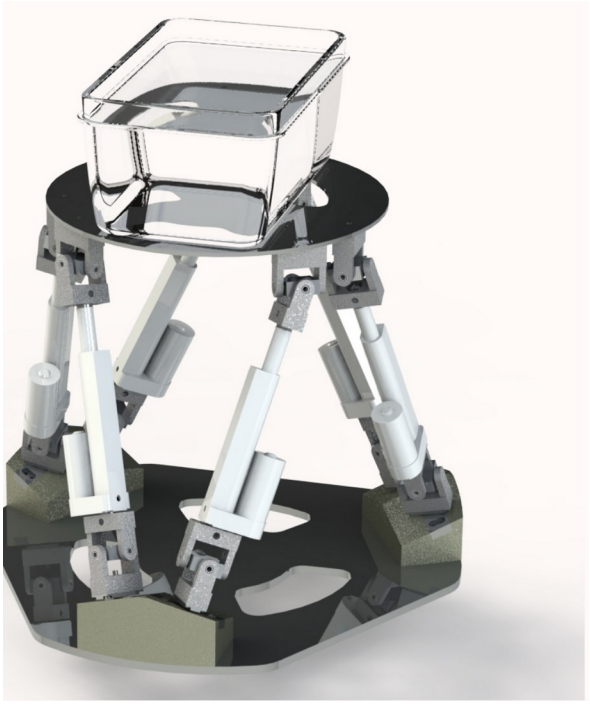


Figure 6. CAD arrangement of the anthropomorphic phantom supported by the moving platform.

- Stage A – CAD/metrology: identify joints and frames; archive CAD geometry and measurement uncertainty.
- Stage B – Actuator commissioning: establish home, individual servo stability, speed limits and encoder calibration.
- Stage C – Full-platform tracking: verify repeated static poses and simple periodic 1-axis waveforms.
- Stage D – Multi-axis tracking: test combined translations/rotations, detect singularity margin and saturation.
- Stage E – Phantom tracking: measure independent phantom surface and internal-target marker response.
- Stage F – Integration: quantify stationary versus moving target needle-insertion performance.

9. Technical Disclosure

9.1 Technical field

The proposed subject matter lies in parallel kinematic motion platforms, programmable phantom-based respiratory simulation, and benchtop assessment of image-guided or robot-assisted tools operating relative to moving anatomical targets.

9.2 Problem addressed

Many bench experiments treat an anatomical target as fixed, obscuring target-position changes associated with respiration. A programmable motion platform can expose instrument control and targeting algorithms to repeatable prescribed movement. The specific engineering challenge is to coordinate six linear axes beneath a payload in a compact geometry while managing joint articulation, actuator workspace and reference-signal fidelity.

9.3 Candidate differentiating construction features

Subject to novelty searching and verification of physical drawings, potentially claim-relevant details include: the particular arrangement of tri-axis-inclined actuator wedges, the internal construction of the bearing-supported custom universal joints, the phantom-specific top plate/interface, calibrated pose-to-leg command generation with combined hardware limit enforcement, and a repeatable target-calibration methodology. These are engineering disclosure candidates, not assertions of patentability. Fundamental 6-DOF Stewart-platform respiratory simulation and robotic needle compensation are documented in the prior literature [5,6].

9.4 Technical Features

- An apparatus comprising a stationary base, a moveable phantom-support plate and six independently actuated length-variable members connected through angle-accommodating joints, wherein commanded plate poses are derived from a reference respiratory motion signal.
- A mechanically defined joint-and-wedge interface for positioning the six members around the plate perimeter, subject to specific as-built geometric features to be added from manufacturing drawings.
- A control method converting a calibrated multidimensional respiratory trajectory into simultaneous actuator-length references while verifying joint limits and workspace feasibility.
- A phantom simulation method measuring both commanded platform pose and internal target motion to assess motion-compensation approaches during bench interventions.

10. Discussion, limitations and next steps

The mechanical contribution is a fabricated six-actuator, 341-component Stewart-platform prototype incorporating custom universal joints and angled mounting interfaces. The derived kinematics support commanded pose-to-leg-length conversion, while published respiratory-motion studies motivate asymmetric and nonstationary input profiles. Full quantitative assessment requires calibrated anchor geometry, experimentally characterized actuator behavior and synchronized reference displacement measurements.

Important unresolved tasks are to freeze the actual anchor geometry, measure full joint limits, document servo electronics and actuator characteristics, establish real-time motion feedback, obtain independent pose measurements, and characterize how movement of the upper plate transfers to the internal target inside the selected phantom. Asymmetric respiratory reference functions are appropriate for commissioning, but patient-specific tests require time-stamped data and clinical context. A platform-level validation does not constitute a validated autonomous surgical system.

11. Conclusions

A six-degree-of-freedom Stewart–Gough mechanism has been designed and assembled to support respiratory motion simulation using an anthropomorphic abdominal phantom. The mechanical system combines six linear actuators, custom articulated joints and angled mounting features. The analytical formulation establishes inverse-kinematic and Jacobian relations for trajectory generation. Respiratory reference models provide a basis for future control implementation, while motion-tracking accuracy, repeatability and anatomical target-motion fidelity remain to be quantified in controlled experiments.

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